Decision Trees & Random Forests: The Essential Visuals

20 Key Charts Every Data Scientist Should Know

Methods and Algorithms

Spring 2026

The Machine Learning Pipeline

XKCD #1838 by Randall Munroe (CC BY-NC 2.5)

Learning Objectives

After this lecture, you will be able to:

- **Visualize** how decision trees partition feature space
- **Understand** the bias–variance tradeoff in tree models
- **See** how random forests improve over single trees
- **Interpret** RF models using modern diagnostic tools

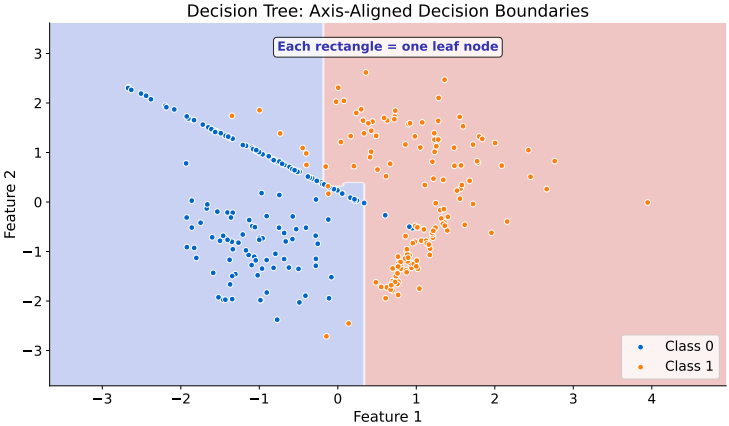
We cover 20 essential visualizations spanning foundations, ensemble mechanics, and model interpretation

Outline

1. Decision Tree Foundations
2. Random Forest Power
3. Interpretation and Evaluation
4. Summary

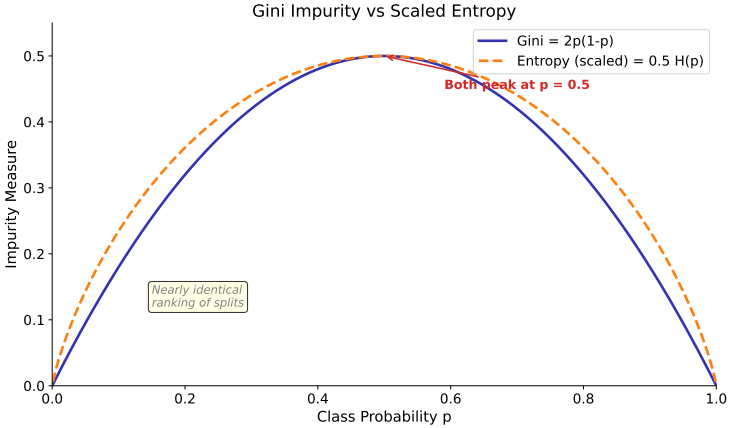
Three parts: Decision Tree Foundations, Random Forest Power, Interpretation and Evaluation

How Does a Decision Tree Classify?



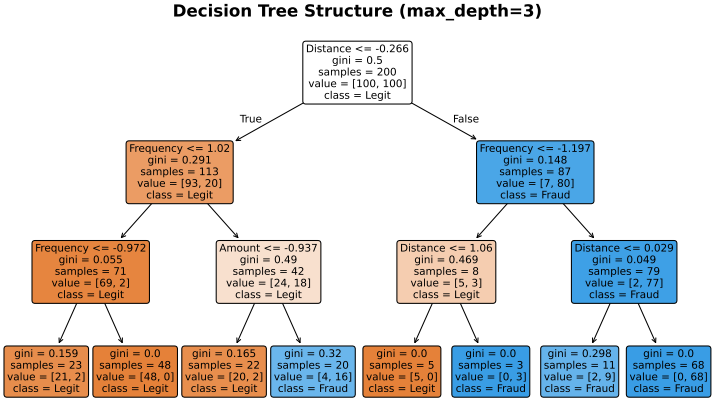
Decision trees create axis-aligned rectangular partitions of the feature space

Impurity Measures: Gini vs Entropy



Both measures reach maximum at $p = 0.5$ (maximum uncertainty) and minimum at $p = 0$ or $p = 1$ (pure nodes)

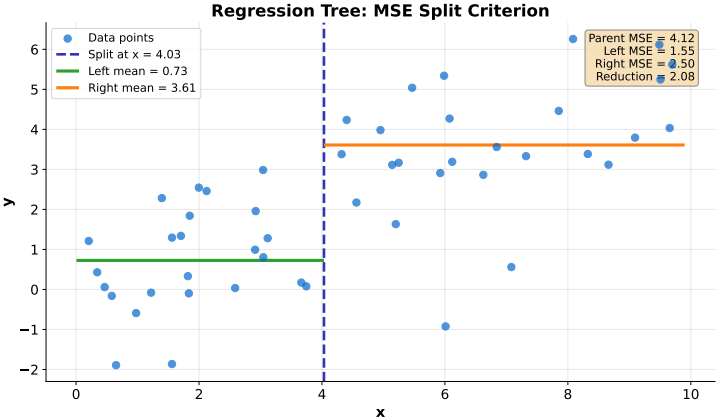
Inside the Tree: Nodes, Splits, and Leaves



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L04_Random_Forests/top10_05_tree_structure

Each node shows the split rule, impurity measure, sample count, and class distribution

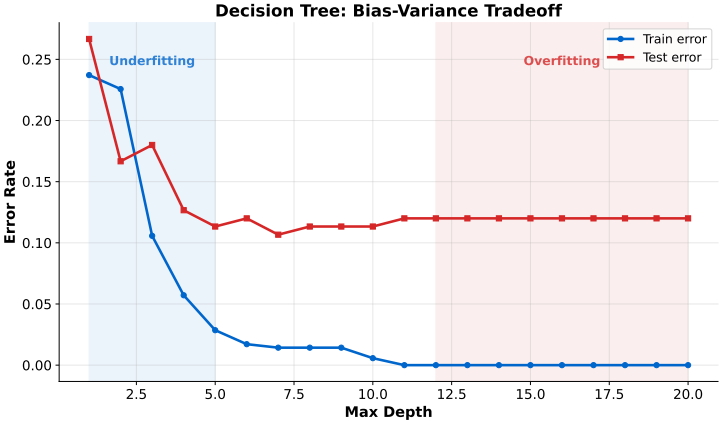
Decision Tree Regression: Piecewise Approximation



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L04_Random_Forests/16_regression_tree_mse

Decision trees approximate continuous functions with piecewise-constant (step) predictions

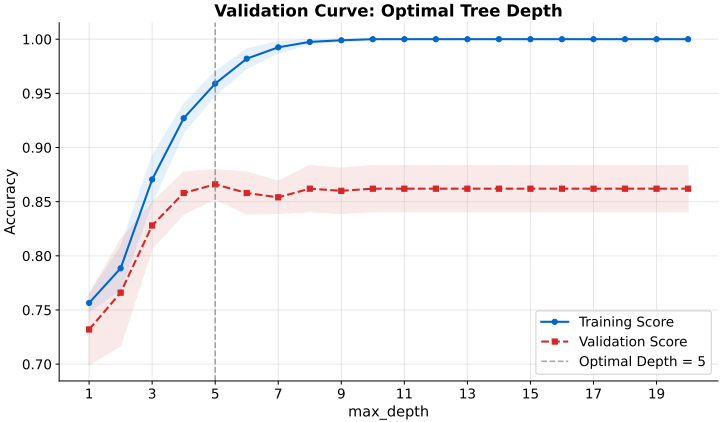
The Core Tradeoff: Tree Depth vs Error



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L04_Random_Forests/17_bias_variance_depth

Shallow trees underfit (high bias); deep trees overfit (high variance). The optimal depth minimizes test error.

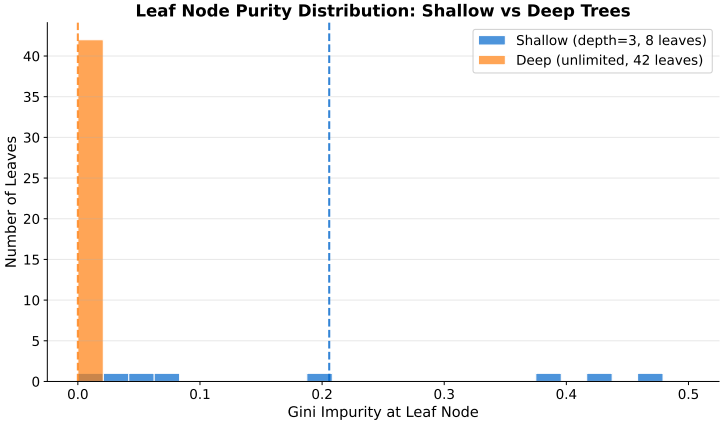
Validation Curve: Finding Optimal Depth



https://github.com/Digital-AI-finance/methods-algorithms/tree/master/slides/L04_Random_Forests/top10_12_validation_curve

Cross-validated train and test scores reveal the sweet spot between underfitting and overfitting

Leaf Node Purity: Shallow vs Deep Trees



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L04_Random_Forests/top10_19_leaf_purity

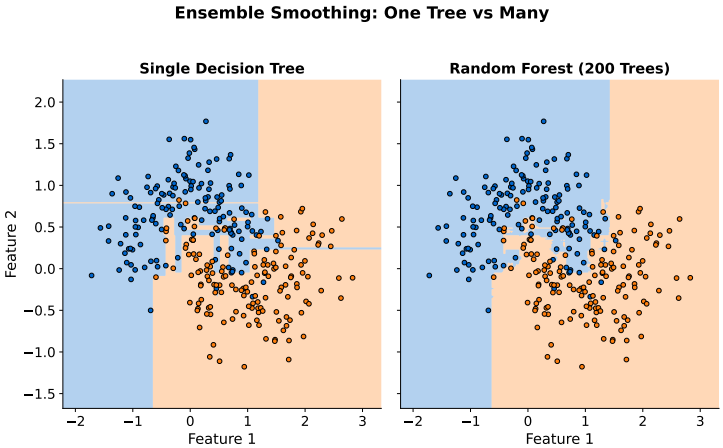
Deeper trees produce purer leaves but risk memorizing noise in the training data

The Problem: Single Trees Are Unstable

- Small changes in data $\rightarrow$ very different tree structures
- Deep trees memorize noise (**overfitting**)
- **Solution:** Combine many diverse trees $\rightarrow$ **Random Forest**

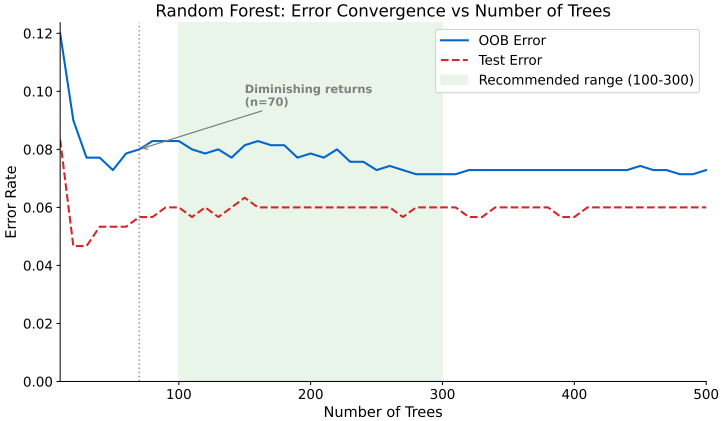
The instability of individual trees is a feature, not a bug—it creates the diversity that makes ensembles work

The Ensemble Effect: Smooth Boundaries



Averaging 200 trees smooths out the jagged decision boundary of a single tree

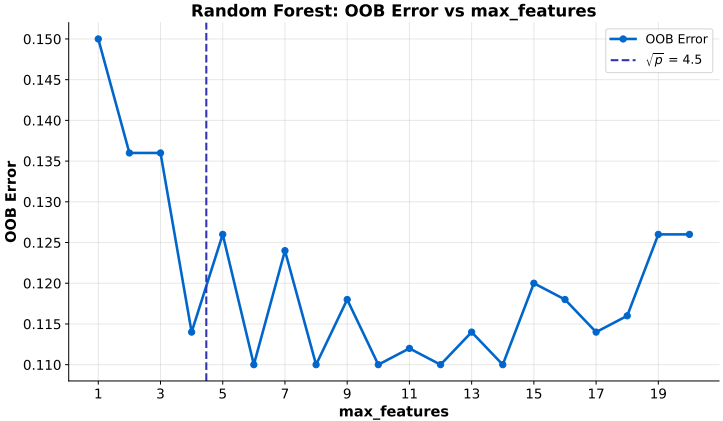
How Many Trees Are Enough?



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L04_Random_Forests/27_rf_ntrees_convergence

OOB error provides free cross-validation; error plateaus after 100–200 trees

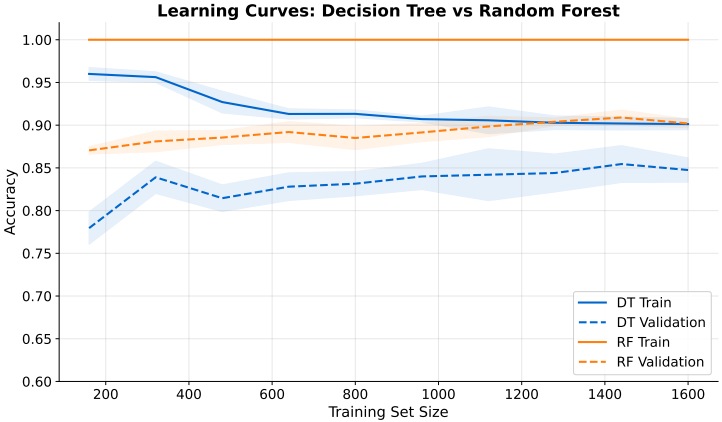
The Key Hyperparameter: max_features



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L04_Random_Forests/18_rf_hyperparameter_maxfeatures

Reducing features per split decorrelates trees, improving ensemble diversity at the cost of individual tree accuracy

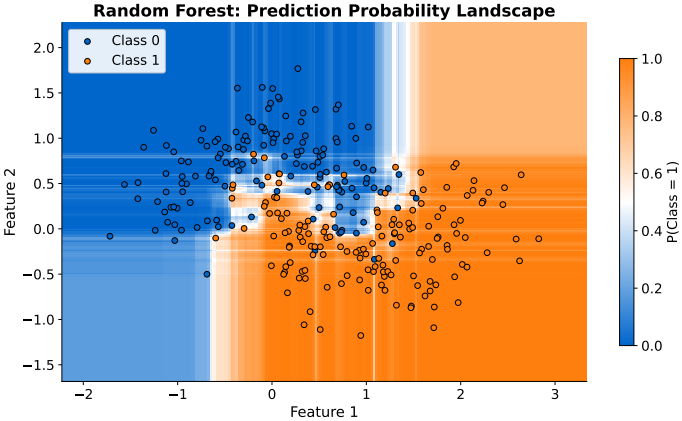
Learning Curves: Decision Tree vs Random Forest



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L04_Random_Forests/top10_11_learning_curve

Random forests generalize better than single trees, especially with limited training data

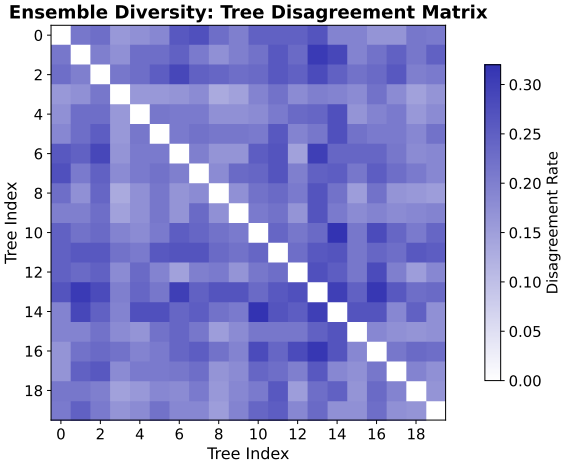
Prediction Probability Landscape



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L04_Random_Forests/top10_16_probability_heatmap

Unlike hard classifications, predict_proba reveals model confidence across the feature space

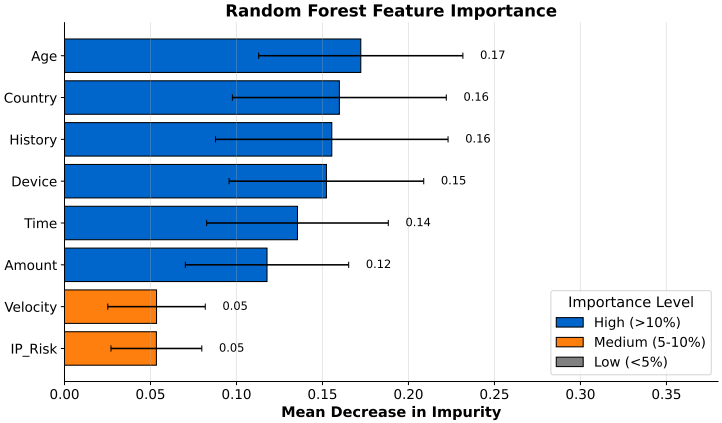
Why Diversity Matters: Tree Disagreement



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L04_Random_Forests/top10_18_ensemble_diversity

Feature randomization ensures trees disagree on individual predictions while agreeing on the overall

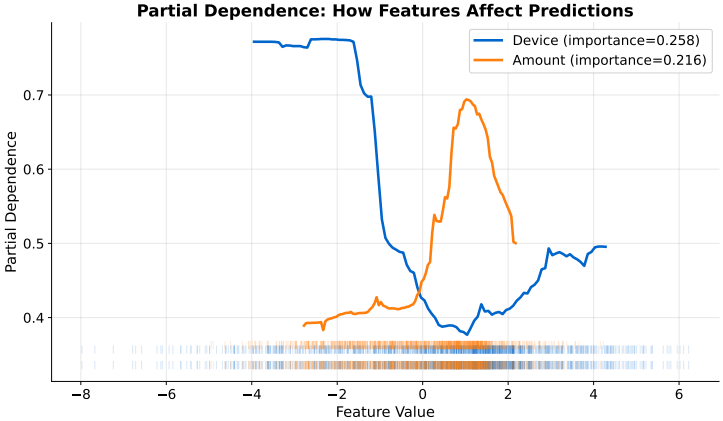
Which Features Matter Most?



https://github.com/Digitar-AI-Finance/methods-algorithms/tree/master/slides/L04_Random_Forests/02_feature_importance

Mean Decrease in Impurity (MDI) ranks features by how much they reduce node impurity across all trees

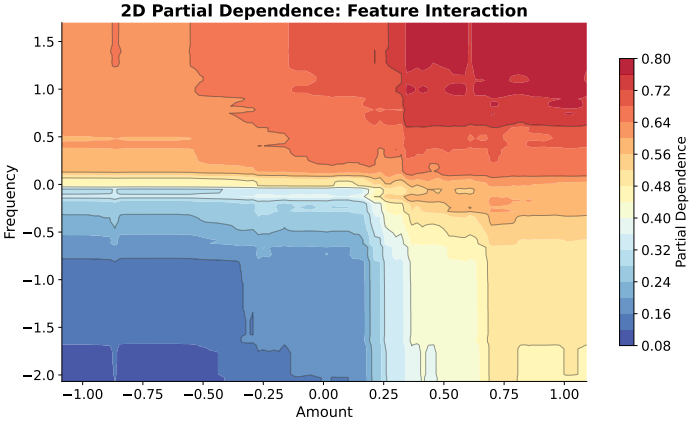
Partial Dependence: Feature Effects



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L04_Random_Forests/top10_10_partial_dependence

PDPs show the marginal effect of a feature on predictions, averaging out all other features

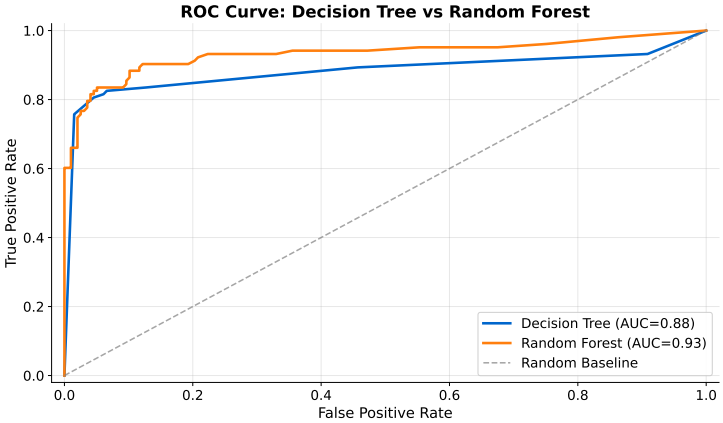
Feature Interaction: 2D Partial Dependence



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L04_Random_Forests/top10_17_2d_pdp_contour

2D PDPs reveal interaction effects between two features that 1D plots cannot capture

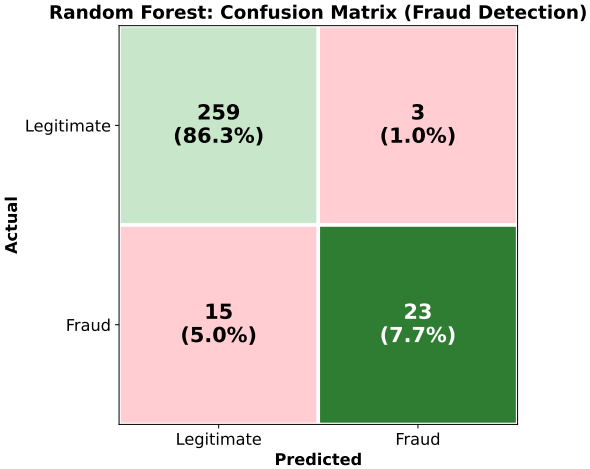
ROC Curves: Single Tree vs Forest



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L04_Random_Forests/top10_13_roc_comparison

Random forests typically achieve higher AUC than single trees due to lower variance in probability estimates

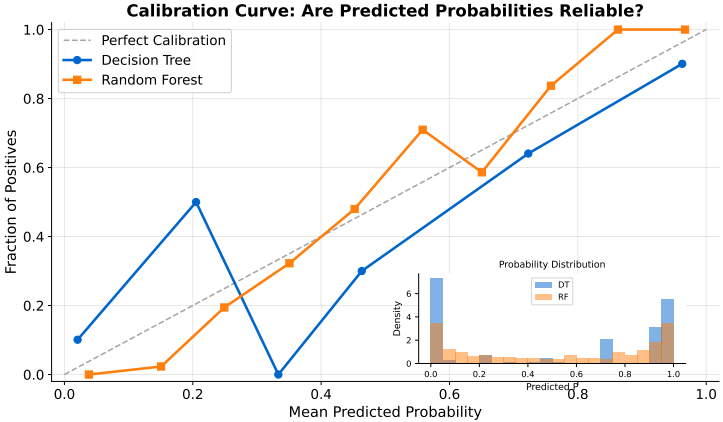
Confusion Matrix: Reading the Results



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L04_Random_Forests/top10_14_confusion_matrix

In fraud detection, minimizing false negatives (missed fraud) is often more important than minimizing false positives

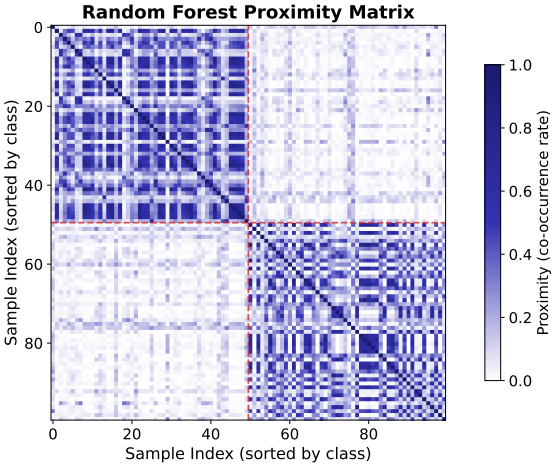
Are the Probabilities Reliable?



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L04_Random_Forests/top10_15_calibration_curve

Well-calibrated models have predicted probabilities that match observed frequencies

Random Forest Proximity: Which Samples Are Similar?



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L04_Random_Forests/top10_20_proximity_matrix

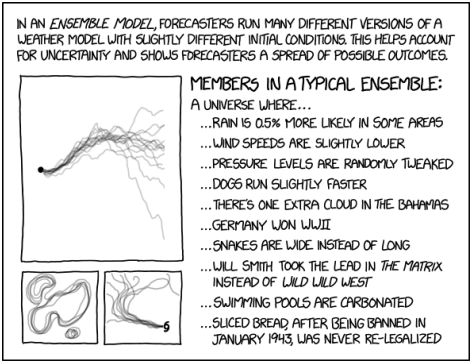
Samples landing in the same leaves are considered similar by the forest

20 Charts You Should Know

1. Decision Boundary
2. Gini & Entropy
3. Tree Structure
4. Regression Fit
5. Bias–Variance
6. Validation Curve
7. Leaf Purity
8. DT vs RF Boundary
9. OOB Convergence
10. max_features
11. Learning Curves
12. Probability Landscape
13. Ensemble Diversity
14. Feature Importance
15. Partial Dependence
16. 2D PDP Interaction
17. ROC Comparison
18. Confusion Matrix
19. Calibration Curve
20. Proximity Matrix

These 20 visualizations cover the foundations, mechanics, and interpretation of tree-based models

The Wisdom of the Crowd



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